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BSc (HONS.) IN COMPUTER SCIENCE AND ENGINEERING
FIRST YEAR, SECOND SEMESTER EXAMINATION, 2020

[According to the New Syllabus]

LINEAR ALGEBRA

Subject Code : CSE-510225

Examination Code : 5612

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

- | | Marks |
|--|-------|
| 4. (a) Define homogeneous and non-homogeneous linear equation.
Reduce the following system of linear equations into echelon form and solve it.
$\begin{aligned}x_1 - x_2 + 2x_3 &= 5 \\ 2x_1 + x_2 - x_3 &= 2 \\ 2x_1 - x_2 - x_3 &= 4 \\ x_1 + 3x_2 + 2x_3 &= 1\end{aligned}$ | 3+5=8 |
| (b) Determine the values of λ such that the following system of linear equations in unknowns x , y and z has (i) a unique solution (ii) no solution (iii) more than one solution :
$\begin{aligned}x + y + \lambda z &= 1 \\ x + \lambda y + z &= \lambda \\ \lambda x + y + z &= \lambda^2\end{aligned}$ | 6 |
| (c) If $2s = a + b + c$ then prove that, | 6 |

$$\begin{vmatrix} a^2 & (s-a)^2 & (s-a)^2 \\ (s-b)^2 & b^2 & (s-b)^2 \\ (s-c)^2 & (s-c)^2 & c^2 \end{vmatrix} = 2s^3 (s-a)(s-b)(s-c).$$

[Please turn over

Marks

2. (a) Define : Singular Matrix, Nilpotent Matrix and Hermitian Matrix. 2×3=6

- (b) If A and B are non-singular matrices, then prove that, 7

$$(AB)^{-1} = B^{-1}A^{-1}$$

- (c) Define the rank of a matrix. Find the rank of the following matrix : 2+5=7

$$\begin{bmatrix} 1 & 2 & 0 & -1 \\ 2 & 6 & -3 & -3 \\ 3 & 10 & -6 & -5 \end{bmatrix}$$

3. (a) If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, prove that, $A^2 = A^{-1}$. 8

- (b) Show that the matrix $A = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ 12

is invertible for all values θ and find A^{-1} .

4. (a) Define linear transformation let $T: \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be given by $T(a, b, c) = (a, b, 0) \forall a, b, c \in \mathbb{R}$. Show that T is a linear transformation. 2+5=7

- (b) Define rank and nullity of a linear transformation. If $T: U \longrightarrow V$ is a linear transformation then show that rank of T + nullity of $T = \dim U$. 2+6=8

- (c) $T: \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear operator defined by 5

$$T(x, y, z) = (x + 2y - z, 2x - y + 4z, 4x + 3y - 2z).$$

Find a basis and the dimension of the (i) $\text{Im } T$ of T

(ii) $\text{Ker } T$ of T .

Marks

5. (a) Define matrix representation of a linear operator. Let $\{u_1, u_2, \dots, u_m\}$ be a basis of U and T be any linear operator of U , then for any vector $W \in U$, 3+7=10

Show that $[T]_u [W]_u = [T(W)]_u$.

- (b) The mapping $T: \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ is defined by

4+6=10

$$T(x, y, z) = (2y + z, x - 4y, 3x).$$

Show that, T is a linear transformation and find the matrix T in the basis $\{u_1 = (1, 1, 1), u_2 = (1, 1, 0), u_3 = (1, 0, 0)\}$.

6. (a) Define eigen values and eigen vectors. State and prove the Cayley-Hamilton theorem. 2+6=8

- (b) Find the eigen values and the corresponding eigen vectors of

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the matrix $A = \begin{vmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 3 \end{vmatrix}$

Also verify Cayley-Hamilton theorem for the matrix A .